MILITARY SPECIFICATION

LAMINATE, ACETAL AND CLOTH, (FOR USE WITH 81MM AMMUNITION)

1. SCOPE

1.1 Scope.-This specification covers a laminate of acetal sheeting bonded by adhesive to cloth for usage as propellant increment bag material for 81MM ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-286	-Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	-Lot Numbering of Ammunition

DRAWINGS

ARMY

9233369	-Charge A, Propellant Increment, M90A1 Mortar Cartridge, 81MM
9233371	-Charge B, Propellant Increment, M90A1 Mortar Cartridge, 81MM

FSC: 1395

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials Publications

D618 -Conditioning Plastics and Electrical Insulating Materials for Testing
D646 -Test for Basis Weight of Paper and Paperboard
D882 -Test for Tensile Properties of Thin Plastic Sheeting

3. REQUIREMENTS

3.1 Materials.-The laminate shall be made from materials in accordance with drawings 9233368 and 9233370.

3.2 Physical Requirements.-The laminate shall comply with the requirements specified in Table I.

TABLE I

Properties	Requirements	Test Methods
Tensile Strength	9.0 lbs min.	4.4.1
Basis Weight (applicable to silk-acetal laminate)	3.0 ozs. per sq. yd., max.	4.4.2
Basis Weight (applicable to acrylic-acetal laminate)	3.5 ozs. per sq. yd., max.	4.4.2
Roll width (Charge A Bag)	2.375 $\pm$. 063 inches	4.4.3
Roll width (Charge B Bag)	2.750 $\pm$ .063 inches	4.4.3

3.3 Strength of Heat Seal.-The laminate, upon being heat sealed to itself on the acetal side, shall withstand application of a static load of 2.5 pounds per lineal inch, minimum, without peel-back of the heat seal and without tensile failure of the area adjacent to the heat seal when tested as specified in 4.4.4.

3.4 First Article Inspection.-A first article sample shall be submitted in accordance with para. 4.2.2. Samples shall be chosen,

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tested and observed for compliance with requirements in accordance with this specification.

3.5 Workmanship.-The laminate shall be free from breaks, cracks, creases, loss of bond or flaws of any kind, and foreign materials.

3.6 Ballistic Comparison Test.-First article samples shall be tested in accordance with para. 4.4.5 to determine compliance with the Ballistic Comparison Test requirements specified therein.

3.7 Vacuum Stability Test.-Samples shall be tested in accordance with para. 4.4.6 to determine compliance with the Vacuum Stability Test requirements specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement of findings on the physical properties of the material in addition to a statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to

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bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.2.2 First article inspection

4.2.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.2.2.2. The first article sample shall consist of one lot of laminated material which has been produced by the contractor and has been manufactured by the same production process, using procedures and equipment which will be used in fulfilling the contract. All materials including the packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects shall not be submitted unless specifically designated by the contracting officer. A first article sample shall also be submitted as

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directed by the contracting officer, whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in the manufacturing process, materials used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2.2 Rejection.-If the laminate fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Sampling Plan.-A group of five (5) rolls of Charge A Bag width and five (5) rolls of Charge B Bag width shall be randomly selected from each lot. A sample shall be removed from each roll, tested and inspected for compliance with the requirements of this specification. No sample shall be taken from the first or last five yards of a roll. The sample shall be conditioned as specified in ASTM Designation D618. The test results from each roll shall be averaged for reporting purposes. Failure of any individual sample to conform to the requirements of this specification shall cause rejection of the entire lot.

4.4 Test methods and procedures

4.4.1 Tensile strength, Major Defect.-The tensile strength shall be determined in accordance with ASTM Designation D882, Method B.

4.4.2 Basis Weight, Major Defect.-The basis weight of the laminate shall be determined in accordance with ASTM Designation D646.

4.4.3 Roll width, Major Defect.-The width of the rolls selected for sampling in accordance with 4.3 shall be gaged for compliance with requirements as specified in Table I.

4.4.4 Strength of Heat Seal, Major Defect.-Two pieces of laminate selected in accordance with 4.3 shall be heat sealed together on the acetal side. The width of the seal shall be 1/32 inch min. and 1/8 inch maximum. The specimen size shall be one inch in width. One leg of the specimen shall be clamped so that the specimen hangs freely. The required load (see 3.3) shall be applied to the free end in such a manner as to avoid impact loading the heat seal. The load shall be allowed to act for one minute after which the seam shall be examined for compliance with the requirement as specified in 3.3.

4.4.5 Ballistic Comparison Test, Critical Defect.- This test will be conducted at a Government Proving Ground utilizing 81MM rounds containing propellant increment bags manufactured

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according to drawings 9233368, 9233369, 9233370, and 9233371. Ballistic test rounds shall be assembled using increment bags made from the first article sample and metal parts from the same lots of metal parts used to assemble the ballistic reference rounds. The increment bags on the reference rounds shall be selected from present production at the time of first article testing. Testing details shall be in accordance with Table II and the following: All groups shall be fired at an elevation of 45 degrees using a 3.209 inch, max. mortar tube at a rate of 12 rounds per minute. The test rounds shall not significantly increase the standard deviation of velocity nor the misfire rate. In addition, the level of velocity shall not decrease as compared to the reference rounds.

TABLE II

ZONE	TEST ROUNDS	REFERENCE ROUND
1	25	25
2	100	100
3	25	25
4	25	25
5	25	25
6	25	25
7	25	25
8	25	25
9	25	25

4.4.6 Vacuum Stability Test, Critical Defect.-The procedure for this test is as specified in Standard MIL-STD-286, Test Method 403.1.2. The following formula shall be used to determine acceptance or rejection:

$$A + B + 5.0 \text{ mls.} > C$$

where:

A = Numbers of mls. of gas evolved by 2.5 gr of M9 propellant in forty hours.

B = Numbers of mls. of gas evolved by 2.5 gr of laminate material in forty hours.

C = Number of mls. of gas evolved by 5.0 gr of a mixture (1:1 ratio) of M9 propellant and finely cut up laminate material.

5. PREPARATION FOR DELIVERY

5.1 Packing

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5.1.1 Level C.-The laminate shall be packed in a manner that will insure safe delivery from the manufacturer's plant to the first receiving activity, and in conformance with applicable carrier regulations.

5.2 Marking.-Interior packages, and exterior containers, shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify:
a. Title, number and date of this inspection.
b. Provisions for submission of first article samples.

6.2 Test results.-A copy of the test results should be forwarded within five days of completion of tests to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-QA-A-A.

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